Comprehensive Review of Finite-control-set Model Predictive Control Methods for Permanent-magnet Synchronous Motors*

Bo Wang, *Shi Jin**, *Peng Sun and Guangwei Liu*

(School of Electrical Engineering, Shenyang University of Technology, Shenyang 110870, China)

Abstract: Permanent-magnet synchronous motors (PMSMs) have become the preferred choice in the industrial sector owing to their high power density, excellent efficiency, and wide speed control range. Over the past decade, advancements in high-performance drives and power electronics have propelled the development of model predictive control (MPC) technology. Among the various approaches, finite-control-set MPC (FCS-MPC) has attracted significant interest for the control of permanent-magnet motors owing to its simple structure and fast response. Three key challenges in FCS-MPC are explored by leveraging recent research advancements: selecting appropriate weighting factors in the cost function, optimizing vector selection, minimizing computation time in multi-vector control, and improving system robustness under parameter mismatches. Additionally, the evolution and application of FCS-MPC in PMSMs are discussed.

Keywords: Permanent-magnet synchronous motor, finite-control-set model predictive control, weighting factor, multi-vector control, parameter mismatch

1 Introduction

Permanent-magnet synchronous motors (PMSMs) are widely employed in fields such as flywheel energy storage [1], servo control systems [2], aerospace [3], and electric vehicles [4] owing to their high reliability, simple design, compact size, wide speed range, high power density, and superior efficiency [5-7]. To satisfy the performance requirements of drive systems across diverse application scenarios, researchers have developed various control strategies, with field oriented control (FOC) [8] and direct torque control (DTC) [9] being the two most prominent. Since the 1960s, model predictive control (MPC), a branch of predictive control, has combined the strengths of FOC and DTC, resulting in extensive research and applications in PMSM drives. Fig. 1 presents the classification of predictive control techniques within the drive-system domain [10]. A defining feature of MPC is the use of system models to predict and manage future changes in variables by selecting optimal control actions based on preestablished criteria [11]. By defining appropriate criteria, MPC can manage multiple critical parameters simultaneously, including the output current, switching frequency, power loss, and motor torque, thus achieving multi-objective optimization [12]. Compared with FOC, MPC is simpler in structure, as it eliminates the need for a proportional-integral (PI) regulator in the current control loop, reducing system overshoot and steady-state errors. Compared with DTC, MPC can integrate torque ripple into its optimization process, which enhances the stability of the control system.

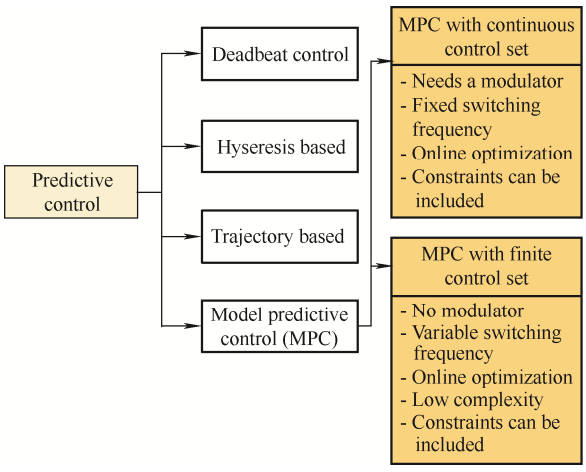


Fig. 1 Classification of predictive control

Manuscript received April 30, 2024; revised May 5, 2024; accepted May 11, 2024.
Date of publication March 31, 2026; date of current version June 15, 2025.
* Corresponding Author, E-mail: wby-js@163.com
* Supported by the Key International Cooperation of National Natural Science Foundation of China (51920105011) and the Applied Basic Research Program of Liaoning Province, China (2023JH2/101300219).
Digital Object Identifier: 10.23919/CJEE.2025.000175
2096-1529 © 2026 China Machinery Industry Information Institute

As shown in Fig. 1, MPC is categorized into continuous control set MPC (CCS-MPC) [13-14] and finite-control-set MPC (FCS-MPC) [15-17] depending on whether pulse width modulation (PWM) is required. CCS-MPC achieves good dynamic performance by designing a cost function that solves constrained quadratic programming problems online, enabling the synthesis of vectors with arbitrary magnitudes and directions to drive the PMSM [18]. However, the CCS-MPC approach is computationally intensive and places high demands on the processor performance. This paper focuses primarily on FCS-MPC, offering a detailed analysis of FCS-MPC strategies in PMSM drive systems, evaluating the challenges encountered in practical applications, and discussing future development directions.

Based on the classification of the control variables, FCS-MPC can be categorized into finite-control-set model predictive current control (FCS-MPCC) [19-20], finite-control-set model predictive torque control (FCS-MPTC) [21-22], and finite-control-set model predictive direct speed control (FCS-MPDSC) [23-24], as shown in Fig. 2.

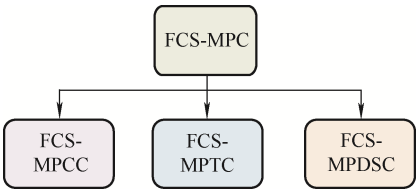


Fig. 2 Classification of FCS-MPC

FCS-MPCC treats the current as the control variable and constructs a cost function based on the current to assess the performance of each voltage vector. The objective is to select a voltage vector that minimizes the current ripple. The objective of FCS-MPTC is to use a cost function to select the voltage vector that optimizes both torque and flux performance, thereby considering both aspects of the control process. However, owing to the different units of torque and flux, appropriate weighting factors must be set to achieve the desired control results. Currently, no mature theory exists for calculating these weighting factors; therefore, they are typically derived from extensive experimental data and experience, making the design process complex and time-consuming. Thus, the accurate determination of weighting factors is a major research focus for FCS-MPTC [25-26]. Similarly, the FCS-MPDSC strategy evaluates both current and speed as control variables within a single cost function. High-performance control with a single closed loop was achieved by removing the PI regulators for both the current and speed loops. However, because the cost function includes variables of different units (current and speed), it encounters a similar challenge in determining the appropriate weighting factors. A comparison of the advantages, disadvantages, and application environments of the three control methods is presented in Tab. 1. Each control method is suitable for different motor control scenarios, and the choice should be based on the actual situation.

Tab. 1 Comparison and analysis of the three control algorithms

Control algorithm	Advantages	Disadvantages	Application environment
FCS-MPCC	Fast response Suitable for high dynamic situations	High model accuracy required Requires precise current estimation	High speed motor drive Applications require high-precision current control
FCS-MPTC	Precise torque control Good response to torque variations	High computational complexity May cause excessive oscillations	Motor speed control Applications require high-precision torque control
FCS-MPDSC	Precise speed control High speed response and low tracking accuracy	Require high system model and speed sensor accuracy	High speed motor drive Applications requiring extremely high speed accuracy

The conventional FCS-MPC typically selects a single optimal voltage vector per cycle, which can result in significant discrepancies between the output and reference voltage vectors, thereby affecting the smooth operation of the system. Two or three vectors are commonly applied within a single cycle to address this [27-28].

However, increasing the number of applied vectors increases the computational burden, posing challenges to the processing power of the system. Reducing the computational load and maintaining a fixed switching frequency have become the key research directions in the study of multi-vector FCS-MPC.

The accurate operation of FCS-MPC depends heavily on the precision of the mathematical model of the controlled system, with parameter mismatches

potentially causing significant fluctuations in the system performance [29-30]. When applied to a PMSM, the *d-q* axis voltage balance equations involve parameters such as the inductance, resistance, and flux linkage of the permanent magnets. In practical applications, these parameters can vary with changes in the operating conditions and environment. For instance, elevated temperatures can increase the stator resistance and lead to the demagnetization of the permanent magnets, resulting in reduced rotor flux linkage and lower *d-q* axis inductance, which can cause torque ripples [31-33]. Additionally, changes in the saturation level of the stator core can alter the stator inductance [34-35]. The uncertainty of motor-parameter variations makes the achievement of optimal control performance difficult for FCS-MPC. Therefore, for enhanced stability of FCS-MPC systems and precise and robust control of PMSMs, accurate motor parameters must be obtained [36-38].

Currently, three primary methods are used to address parameter mismatches in FCS-MPC control systems for permanent-magnet motors: observer-based estimation and compensation [39], motor-parameter identification techniques [40-44], and model optimization [45]. The observer-based method treats parameter mismatches as system disturbances and uses observers to estimate and compensate for these disturbances or extract parameter information from system errors [46-47]. This approach is widely used to enhance parameter robustness; however, it can only estimate the variation in a single parameter at a time. Motor-parameter identification techniques enable the real-time identification of various electrical parameters of PMSMs, such as inductance, resistance, and flux linkage, thereby improving the robustness of the system parameters. However, the simultaneous identification of all parameters requires addressing the rank-deficiency problem. Common solutions include constructing a full-rank equation or using step-by-step identification. The third method, model optimization, reduces the impact of parameter mismatches by constructing incremental models or employing model-free predictive methods [48-49], thereby minimizing the number of parameters in a system. This method is often combined with the first two methods, using observers or identification techniques to obtain information on the remaining parameters, further enhancing system robustness [50].

In summary, this paper provides a systematic review and categorization of the existing problems in FCS-MPC methods for permanent-magnet motors, as illustrated in Fig. 3. Owing to space limitations, all the studies in the field of model control cannot be addressed. Therefore, this discussion focuses on major research achievements over the past decade.

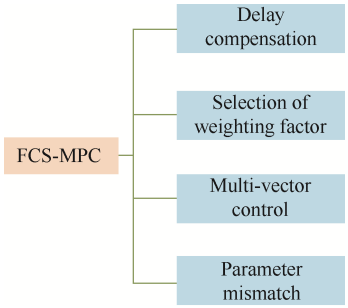


Fig. 3 Main problems with FCS-MPC

This paper is organized as follows. The introduction summarizes the current applications of FCS-MPC in the motor domain, with a focus on permanent-magnet motors. Section 2 details the mathematical model of PMSMs and the fundamentals of the traditional FCS-MPC. Section 3 presents the solutions for determining the weighting factors in the cost function. Section 4 examines the implementation of MPC using double- and triple-vector strategies. Section 5 investigates the methods to ensure the stability of the control system in the presence of parameter mismatches. Sections 3, 4, and 5 build on the application context provided in the introduction. Section 6 discusses the future trends and challenges. Section 7 concludes the paper.

2 Mathematical model of PMSM and core principles of FCS-MPC

2.1 Mathematical model of a surface-mounted PMSM

The mathematical model of a surface-mounted PMSM in the *d-q* rotating reference frame is defined as follows [51]

$$\begin{cases} u_d = R_s i_d + L_d \dfrac{\mathrm{d}i_d}{\mathrm{d}t} - \omega_e L_q i_q \\ u_q = R_s i_q + L_q \dfrac{\mathrm{d}i_q}{\mathrm{d}t} + \omega_e L_d i_d + \omega_e \psi_f \end{cases} \tag{1}$$

where i_d and i_q denote the currents on the d- and q-axis, respectively. u_d and u_q denote the voltages on the d- and q-axis, respectively. L_d and L_q denote the d- and q-axis inductances. R_s denotes the resistance of the stator phase. ω_e is the electronic angular speed of the motor, and ψ_f represents the flux linkage.

$$T_e = \frac{3}{2} p_n i_q \left[\psi_f + (L_d - L_q) i_d\right] \tag{2}$$

$$J \frac{\mathrm{d}\omega_m}{\mathrm{d}t} = T_e - T_L - B_m \omega_m \tag{3}$$

where T_e is the electromagnetic torque, p_n is the number of pole pairs in the motor, J is the system inertia, B_m is the system friction coefficient, and T_L and ω_m represent load torque and mechanical angular speed, respectively.

2.2 Core principles of FCS-MPC

For current control, Eq. (1) can be reformulated as

$$\begin{cases} L_d \dfrac{\mathrm{d}i_d}{\mathrm{d}t} = -R_s i_d + \omega_e L_q i_q + u_d \\ L_q \dfrac{\mathrm{d}i_q}{\mathrm{d}t} = -R_s i_q - \omega_e L_d i_d + u_q - \omega_e \psi_f \end{cases} \tag{4}$$

To predict the future state and determine the current at the next time step, the continuous model must be discretized. Applying forward Euler discretization with a sampling period T_s to Eq. (4) yields the predicted current equation, which can be expressed as [52]

$$\begin{cases} i_d(k+1) = \left(1 - \dfrac{T_s R_s}{L_d}\right) i_d(k) + \dfrac{T_s \omega_e L_q}{L_d} i_q(k) + \dfrac{T_s}{L_d} u_d(k) \\ i_q(k+1) = \left(1 - \dfrac{T_s R_s}{L_q}\right) i_q(k) - \dfrac{T_s \omega_e L_d}{L_q} i_d(k) + \dfrac{T_s}{L_q} u_q(k) - \dfrac{T_s}{L_q} \omega_e(k) \psi_f \end{cases} \tag{5}$$

where k and k+1 represent the values at the current and subsequent moments, respectively.

The control block diagrams for FCS-MPCC, FCS-MPTC, and FCS-MPDSC are shown in Figs. 4-6.

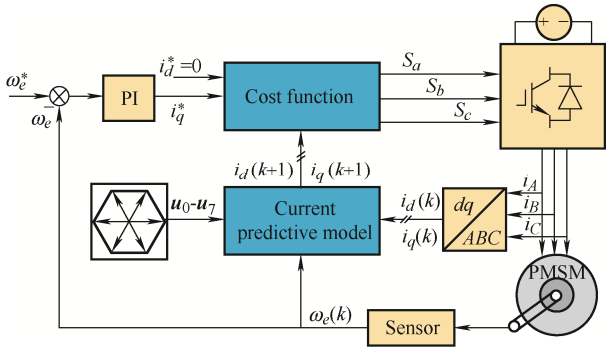


Fig. 4 Control block diagram of FCS-MPCC

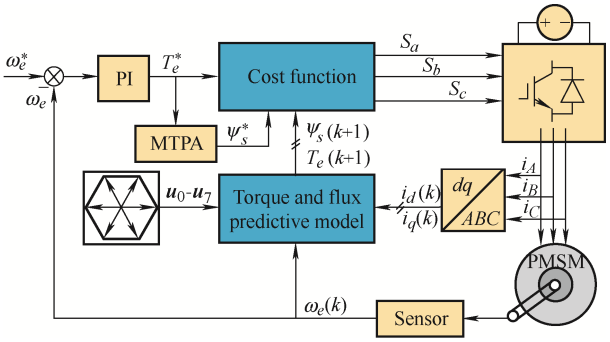


Fig. 5 Control block diagram of FCS-MPTC

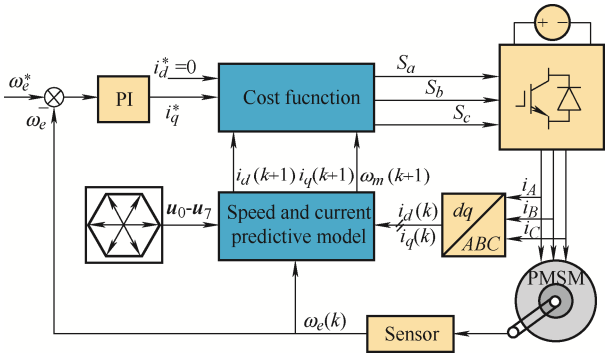


Fig. 6 Control block diagram of FCS-MPDSC

The cost functions for the three variations of FCS-MPC are listed in Tab. 2.

Tab. 2 Three cost function types in FCS-MPC

Algorithm	Cost function	Need weighting factor
FCS-MPCC	$g = \left\|i_d^* - i_d(k+1)\right\| + \left\|i_q^* - i_q(k+1)\right\|$	N
FCS-MPTC	$g = \left\|T_e^* - T_e(k+1)\right\| + \lambda \left\|\psi_s^* + \psi_s(k+1)\right\|$	Y
FCS-MPDSC	$g = \lambda_m \left\|\omega_m^* - \omega_m(k+1)\right\| + \lambda_i \left\|i_d^* + i_d(k+1)\right\| + i_{lim}(k+1)$	Y

2.3 One-step delay compensation

In engineering applications, digital drive systems typically experience a one-step delay, where the optimal solution calculated in the current cycle is delayed until the next cycle for implementation, which can adversely affect system stability. Hence, a

compensation for the one-step delay is required. Based on the current prediction model expressed in Eq. (5), the compensated current for the one-step delay can be derived as follows

$$\begin{cases} i_d^p(k+1)=\left(1-\dfrac{T_sR}{L}\right)i_d(k)+T_s\omega_e i_q(k)+\dfrac{T_s}{L}u_d(k) \\ i_d^p(k+1)=\left(1-\dfrac{T_sR}{L}\right)i_q(k)-T_s\omega_e i_d(k)+ \\ \qquad \dfrac{T_s}{L}u_q(k)-\dfrac{T_s}{L}\omega_e\psi_f \end{cases} \tag{6}$$

where $i_d^p(k+1)$ and $i_q^p(k+1)$ denote the one-step delay compensation currents on the *d-q* axis. Substituting the compensated current into Eq. (5) yields the final predicted current as

$$\begin{cases} i_d(k+2)=\left(1-\dfrac{T_sR}{L}\right)i_d^p(k+1)+ \\ \qquad T_s\omega_e i_q(k)+\dfrac{T_s}{L}u_d(k+1) \\ i_q(k+2)=\left(1-\dfrac{T_sR}{L}\right)i_d^p(k+1)-T_s\omega_e i_d(k+1)+ \\ \qquad \dfrac{T_s}{L}u_q(k+1)-\dfrac{T_s}{L}\omega_e\psi_f \end{cases} \tag{7}$$

Therefore, the cost function can be designed as

$$g=\left|i_d^*-i_d(k+2)\right|+\left|i_q^*-i_q(k+2)\right| \tag{8}$$

3 Computation and elimination of weighting factors

As mentioned in the Section 2, the FCS-MPC system for a PMSM enables the simultaneous control of multiple variables, such as flux linkage and torque or current and speed, while minimizing the cost function. This necessitates solving the problem of determining the appropriate values for the weighting factors when constructing a cost function. In conventional approaches, the values of the weighting factors are typically determined empirically without a clear theoretical basis, and the process requires numerous experiments, which are time-consuming and lack precision. Consequently, designing optimal weighting factors has become a critical focus in MPC research. The optimization of the weighting factors can be broadly classified into numerical computation methods [53-57] and weighting-factor-free approaches [21, 58-64].

3.1 Numerical computation methods

The essence of the numerical computation method for determining the weighting factors is parameter optimization. This process uses modern algorithmic techniques to iteratively compute and determine the optimal solution, either online or offline.

In Ref. [53], based on an artificial neural network, a genetic algorithm was employed as a backpropagation method to derive optimal weighting factors through offline training. This approach was developed based on a PMSM drive system and can be generalized for optimizing the performance of weighting factors in other types of FCS-MPC. The offline training method does not increase the computational complexity of the control system. However, when a system encounters unforeseen disturbances, measures must be taken to enhance its robustness and disturbance-rejection capabilities. In Ref. [54], a fuzzy predictive DTC strategy was adopted, incorporating fuzzy logic into the optimization of weighting factors to dynamically determine their optimal values, thus enabling the selection of the optimal switching vector. Compared with the conventional MPTC, the introduction of dynamic weighting factors through the fuzzy strategy improves the transient torque response speed, reduces the torque ripple, and enhances the speed tracking accuracy. In Ref. [55], a neural network (NN)-based method was used to design the weighting factors for the cost function by incorporating the flux linkage reference value into the optimization process to evaluate how the weighting factors depend on different load conditions. However, because this method also uses offline computation, improvements are required to enhance the disturbance-rejection capabilities of the system. In Refs. [56-57], a particle swarm optimization algorithm was applied for the self-tuning of weighting factors, aiming to reduce the torque ripple, lower the total harmonic distortion (THD) of the current, and decrease the switching frequency.

3.2 Weighting-factor-free approaches

In addition to using numerical computation methods to obtain the optimal weighting factors, the simultaneous control of multiple variables can be achieved by eliminating the weighting factors. In Ref. [58], the conventional MPTC cost function using weighting factors was transformed into a single-objective optimization approach: The torque and flux linkage

cost functions are predicted sequentially. First, the voltage vector that minimizes the torque cost function is selected as a candidate vector for the next prediction step, and then the voltage vector that minimizes the flux linkage cost function is selected from these candidates as the optimal voltage vector. This cascaded structure of the two cost functions effectively eliminates the influence of the weighting factors. However, the two-step prediction process is time-consuming, placing a significant burden on the computational capacity of the controller, making it difficult to apply to high-speed, high-sampling-frequency motor control systems. In Ref. [59], a two-stage serial model predictive approach was employed, with the key difference being that the cost function was constructed based on the instantaneous power theory. The proposed method converts the torque and flux linkage into active and reactive torques, effectively eliminating the influence of the weighting factors. Additionally, the prediction horizon was extended to two periods, and the cost functions for both periods were predicted sequentially to create a serial control structure. In Ref. [60], the two serial cost functions were modified to operate in parallel, which not only resolves the problem of weighting factors in conventional MPTC but also overcomes the limitations of serial cost function operations. However, both serial and parallel approaches rely on separate cost functions for the flux linkage and torque, making them less intuitive and straightforward than using a single cost function. In Ref. [61], a state variable normalization method that unifies the current and speed into physical quantities of the same dimension was proposed. By analyzing the sensitivity of the control variables to changes in voltage, this method achieves unified control of current and speed. The main advantage of this algorithm is that it enables the control of different dimensional physical quantities using a single cost function without requiring weighting factors, thereby providing a reference for future extensions that integrate multiple control variables of different dimensions into a single cost function. In Refs. [21, 62], a cost function was reconstructed using the tracking error of the voltage vector, replacing the torque and flux linkage terms in the traditional cost function, thereby eliminating the need for weighting factors. In Ref. [63], a duty-cycle modulation strategy was employed to achieve torque control objectives, enabling the cost function to focus solely on the flux linkage and completely eliminate the weighting factors used in traditional cost functions. In Ref. [64], a preset torque boundary ensured that all valid candidate voltages provided the predicted torque to this boundary, thereby maintaining a constant difference between the torque reference and the predicted values. Consequently, the cost function is no longer required to evaluate the torque and includes only the flux linkage as a single variable, effectively eliminating the need for weighting factors. However, because this method maintains a constant torque error, incorrect selection of the torque boundary can result in significant deviations in the overall control system.

In summary, both approaches for solving the problem of weighting factors have shown good performance in PMSM drive systems. However, numerical computation methods require the use of various advanced algorithms to estimate weighting factor values. Balancing the reduction in algorithmic complexity with the accuracy of the estimated values remains an important direction for future research. However, methods that eliminate the weighting factors alter the structure of the cost function, and the need for multiple calculations can slow the response of the system, restricting its application in high-speed permanent-magnet motor drive systems.

A summary of the strengths and weaknesses of the aforementioned weighting factor design methods is given in Tab. 3.

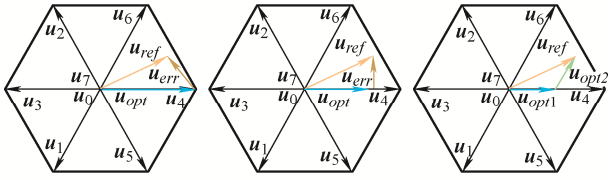
Tab. 3 Comparison and analysis of the design method

Design method	Strengths	Weaknesses
Empirical method	Simple and fast Suitable for familiar system	Require strong system experience May not guarantee optimal performance
Numerical method	Provide precise settings Theoretically achieve optimal performance	Optimization process may take a long time Depend on initial guess
Weighting factor free	Avoid human bias. Improved robustness	Not suitable for all control systems Require a deep system understanding

4 Multi-vector control

As shown in Fig. 7, the conventional FCS-MPC for permanent-magnet motors uses only one of the eight

basic voltage vectors per cycle, with a fixed magnitude and phase angle. This results in a significant discrepancy between the output and reference voltage vectors, leading to large current ripples and torque pulsations in the system. In the double-vector approach, when the second candidate vector is a zero vector, the output voltage vector can have an adjustable magnitude but its phase remains fixed. When using three-vector modulation, both the magnitude and phase angle of the output voltage vector can be adjusted, reducing the error between the output and reference voltage vectors to a minimum.



(a) Single vector (b) Double vector (c) Three vector

Fig. 7 Error between output and reference vectors

4.1 Double-vector control

In the double-vector control, each control cycle can apply two nonzero vectors, one nonzero vector, or one zero vector. The selection of vectors and the distribution of their application times are central to research on two-vector control.

In Refs. [27, 65-68], detailed research was conducted on the selection of the second vector and the calculation of the application time for each vector in FCS-MPC systems. In Ref. [27], a strategy was proposed to use a nonzero vector instead of a zero vector, addressing the limitations in torque control performance caused by the conventional approach, where the second vector is always a zero vector. By analyzing the variations in the torque and flux linkage of the PMSM, the differences between them were identified, leading to the development of a selection table for the second vector that identifies the optimal vector combination to minimize torque ripple. Additionally, the duty-cycle calculation method was optimized to reduce the computational load of the control system. In Ref. [65], building on the conventional double-vector approach, the final vector from the previous control cycle was used in the current cycle, whereas two control vectors are introduced within a single control cycle to reduce the switching frequency and enhance steady-state control performance. Ref. [66] presented a method that incorporates dead time into the control cycle, equating it to a voltage vector with adjustable action time. This leads to a double-vector MPC with variable dead time, which not only reduces the current harmonic distortion but also lowers the switching frequency. In Ref. [67], optimizations were performed for both the selection of the optimal voltage vector and determination of the vector duration. The process was simplified by selecting the voltage vector closest to the reference voltage vector and eliminating the need to calculate the current slopes when determining the vector duration. In Ref. [68], a double-vector control system with a fixed switching frequency was proposed to address the high computational demands of double-vector systems. This method optimizes the number of candidate voltage vectors and maintains a fixed switching frequency. Compared with conventional methods, this approach is particularly effective in maintaining a stable switching frequency and reducing computational complexity while improving the stability and reliability of the permanent-magnet motor drive system.

4.2 Three-vector control

Extending from the double-vector MPC, the three-vector MPC selects three voltage vectors in a single control cycle—two nonzero vectors and one zero vector—to synthesize the optimal voltage vector. Compared with the double-vector approach, this method further reduces the current distortion and torque ripple. However, it makes the calculation of the vector application time more complex. As indicated by Eq. (4), the *d*-*q* axis current slope under the influence of the zero vector can be expressed as

$$\begin{cases} S_{d0}=\dfrac{\mathrm{d}i_d}{\mathrm{d}t}\bigg|_{u_d=0}=\dfrac{1}{L_s}\left(-R_s i_d+\omega_e L_s i_q\right) \\ S_{q0}=\dfrac{\mathrm{d}i_q}{\mathrm{d}t}\bigg|_{u_q=0}=\dfrac{1}{L_s}\left(-R_s i_q-\omega_e L_s i_d-\omega_e \psi_f\right) \end{cases} \tag{9}$$

Hence, the *d*-*q* axis current slope when two nonzero vectors are applied can be calculated as

$$\begin{cases} S_{d1}=\dfrac{\mathrm{d}i_d}{\mathrm{d}t}\bigg|_{\boldsymbol{u}_d=\boldsymbol{u}_{d1}}=S_{d0}+\dfrac{\boldsymbol{u}_{d1}}{L} \\ S_{q1}=\dfrac{\mathrm{d}i_q}{\mathrm{d}t}\bigg|_{\boldsymbol{u}_q=\boldsymbol{u}_{q1}}=S_{q0}+\dfrac{\boldsymbol{u}_{q1}}{L} \end{cases} \tag{10}$$

$$\begin{cases} S_{d2}=\left.\dfrac{\mathrm{d}i_d}{\mathrm{d}t}\right|_{\boldsymbol{u}_d=\boldsymbol{u}_{d2}}=S_{d0}+\dfrac{\boldsymbol{u}_{d2}}{L} \\ S_{q2}=\left.\dfrac{\mathrm{d}i_q}{\mathrm{d}t}\right|_{\boldsymbol{u}_q=\boldsymbol{u}_{q2}}=S_{q0}+\dfrac{\boldsymbol{u}_{q2}}{L} \end{cases} \tag{11}$$

where $\boldsymbol{u}_{d1}$, $\boldsymbol{u}_{q1}$, $\boldsymbol{u}_{d2}$, and $\boldsymbol{u}_{q2}$ are nonzero voltage vectors applied to the *d*-*q* axis, respectively. S_{q0}, S_{d0}, S_{q1}, S_{d1}, S_{q2}, and S_{d2} are the *d*-*q* axis current slopes under the application of zero and nonzero vectors, respectively.

Based on deadbeat control theory, the predictive current formula for the *d*-*q* axis is derived as

$$\begin{cases} i_d^*=i_d(k+1)=i_d(k)+S_{d0}t_0+S_{d1}t_1+S_{d2}t_2 \\ i_q^*=i_q(k+1)=i_q(k)+S_{q0}t_0+S_{q1}t_1+S_{q2}t_2 \end{cases} \tag{12}$$

where t_0, t_1, and t_2 denote the application times for one zero vector and two nonzero vectors, respectively.

From the above equations, the application times for each vector can be determined as

$$t_1=\frac{1}{M}\Big[\left(i_d^*-i_d(k)\right)\left(S_{q2}-S_{q0}\right)+\left(i_q^*-i_q(k)\right)\left(S_{d0}-S_{d2}\right)+t_s\left(S_{q0}S_{d2}-S_{q2}S_{d0}\right)\Big] \tag{13}$$

$$t_2=\frac{1}{-M}\Big[\left(i_d^*-i_d(k)\right)\left(S_{q1}-S_{q0}\right)+\left(i_q^*-i_q(k)\right)\left(S_{d0}-S_{d1}\right)+t_s\left(S_{q0}S_{d1}-S_{q1}S_{d0}\right)\Big] \tag{14}$$

$$M=S_{q0}S_{d2}+S_{q1}S_{d0}+S_{q2}S_{d1}-S_{q1}S_{d2}-S_{q2}S_{d0}-S_{q0}S_{d1} \tag{15}$$

$$t_0=t_s-t_1-t_2 \tag{16}$$

Compared with double-vector control, the addition of an extra vector in a single cycle increases the computational complexity. Efficiently selecting the output voltage vector while reducing computational complexity and saving computation time is a central focus in the research on the three-vector FCS-MPC.

In Ref. [28], a method was proposed to optimize the computation time of three-vector control using a cost function to evaluate three nonadjacent voltage vectors. This approach quickly selects two candidate nonzero vectors, which are then combined with a zero vector to form the final voltage vector. In Ref. [69], a simplified MPC method that uses virtual voltage vectors with control factors was introduced. This method generates virtual vectors that replicate the control effects of traditional vectors while significantly reducing computational costs. In Ref. [70], a time-varying control period method that eliminates the use of the zero vector in three-vector algorithms was used, enabling double-vector control to achieve the performance level of three-vector control. This method effectively reduces the switching frequency and suppresses the common-mode voltage. In Ref. [71], a three-vector FCS-MPTC approach for permanent-magnet motors was developed using a fast selection table. This method reduces the complexity of the algorithm by allowing quick vector selection while synthesizing output voltage vectors with adjustable direction and magnitude, thereby improving the traditional three-vector MPC. It balances the torque performance with the flux linkage fluctuations. In Ref. [72], three-vector MPC was applied to ultra-high-speed PMSM control systems to address steady-state current tracking errors and high computational demands using a double-loop control strategy that eliminates steady-state current prediction errors. Compared with conventional algorithms, this approach reduces complexity, enhances computational efficiency, and optimizes the distribution of non-pure delay errors in the *d*-*q* axis, providing insights for the application of MPC in high-speed PMSMs. Ref. [73] proposed a novel switching table for a three-vector FCS-MPTC that minimizes the computational complexity. It employs deadbeat control principles for torque and flux linkage to calculate duty cycles, effectively reducing current distortion while maintaining dynamic response and steady-state performance.

Tab. 4 compares and summarizes the multi-vector FCS-MPC methods.

Tab. 4 Comparison of multi-vector FCS-MPC

Item	Conventional FCS-MPC	Double-vector FCS-MPC	Three-vector FCS-MPC
Output voltage amplitude	Fixed	Limited fixed	Variable
Output voltage phase angle	Fixed	Fixed	Variable
Dynamic performance	Similar		
Steady-state performance	Improve from left to right		
Computational burden	An increase from left to right		

5 Robustness analysis under parameter mismatch

FCS-MPC delivers a fast response and is highly capable of addressing nonlinear, multi-objective, and multi-constraint problems. However, its performance critically depends on the accuracy of the mathematical model. The primary parameters affecting the model precision include the stator resistance, *d-q* axis inductance, and permanent-magnet flux linkage. This section analyzes the effects of parameter mismatch on the system. The current equation is reformulated as

$$\begin{cases} i_{dm}(k+1)=\left(1-\dfrac{T_sR_0}{L_0}\right)i_d(k)+T_s\omega_e i_q(k)+\dfrac{T_s}{L_0}u_d(k) \\ i_{qm}(k+1)=\left(1-\dfrac{T_sR_0}{L_0}\right)i_q(k)-T_s\omega_e i_d(k)+ \\ \qquad \dfrac{T_s}{L_0}u_q(k)-\dfrac{T_s}{L_0}\omega_e\psi_{f0} \end{cases} \tag{17}$$

where R_0, L_0, and ψ_{f0} denote the resistance, inductance, and flux linkage under parameter mismatch, respectively. $i_{dm}(k+1)$ and $i_{qm}(k+1)$ are the actual predicted *d-q* axis currents under a parameter mismatch. The error between the theoretical current in Eq. (5) and the predicted current in Eq. (17) is expressed as

$$\begin{cases} E_d=\dfrac{T_sR_0\Delta L-T_s\Delta RL_0}{L_0(L_0+\Delta L)}i_d(k)-\dfrac{T_s\Delta L}{L_0(L_0+\Delta L)}u_d(k) \\ E_q=\dfrac{T_sR_0\Delta L-T_s\Delta RL_0}{L_0(L_0+\Delta L)}i_q(k)-\dfrac{T_s\Delta L}{L_0(L_0+\Delta L)}u_q(k)+ \\ \qquad \dfrac{\Delta L\psi_{f0}-L_0\Delta\psi_f}{L_0(L_0+\Delta L)}T_s\omega_e \end{cases} \tag{18}$$

Eq. (18) indicates that only T_sR/L is related to the resistance parameter, and because $T_sR/L \ll 1$, the error caused by the resistance has a negligible impact on the prediction results. The flux linkage error appears only in the *q*-axis current and has minimal effect on the *d*-axis current. However, when the inductance parameter is mismatched, both *d*-axis and *q*-axis currents undergo significant fluctuations. Therefore, the accurate acquisition of motor parameters, particularly the inductance parameter, is crucial for enhancing the performance of the control system. The simulation results shown in Figs. 8 and 9 validate the theoretical analysis results. The motor parameters used in the simulation are listed in Tab. 5.

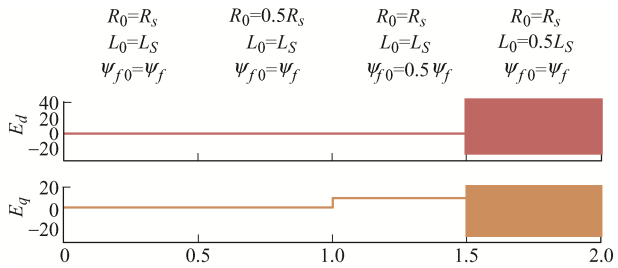


Fig. 8 Error waveform under parameters decrease

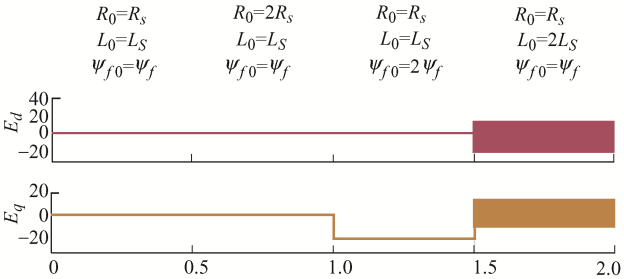


Fig. 9 Error waveform under parameter increase

Tab. 5 Parameter of PMSM

Parameter	Value
Stator resistance/Ω	0.006
Stator inductance/H	0.000 106
Flux linkage/Wb	0.067
Number of pole pairs	2
Inertia/(kg • m^2)	0.009 4
Rated speed/(r/min)	15 000

The simulation results indicate that mismatches in the resistance parameter have a slight effect on E_d and E_q. When the flux linkage parameter is mismatched, E_q is significantly affected, whereas E_d is barely affected. By contrast, mismatches in the inductance parameter have a significant impact on both E_d and E_q. These simulation results are consistent with the theoretical analysis.

The following three approaches are commonly used to resolve the parameter-mismatch problem in the FCS-MPC system for PMSMs: ① observer-based estimation and compensation, ② online parameter identification, and ③ model optimization.

5.1 Observer-based estimation and compensation

PMSMs are subjected to various disturbances during operation, such as parameter variations, load disturbances, and inverter dead-time effects [74]. The suppression of such disturbances and enhancement of

system robustness have been major research focuses. This paper specifically addresses control system instability resulting from motor-parameter mismatches while paying less attention to system problems caused by other external disturbances.

As parameter variations are unpredictable, they can be considered as system disturbances. The use of different types of observers, such as the sliding mode observer (SMO) [75-77], Luenberger observer (LO) [78-80], and disturbance observer (DO) [81-83] to compensate for these disturbances or extract parameter information can effectively mitigate the adverse effects caused by parameter mismatches. Tab. 6 summarizes the key characteristics of various observers.

Tab. 6 Comparison and analysis of observer methods

Observer method	Advantages	Disadvantages
SMO	Strong robustness against parameter and external disturbances	Chattering problem
LO	Simple structure, intuitive design	Poor robustness against model uncertainties
DO	Effective handling of external disturbances	Perform poorly when the system state is not fully understood

5.2 Online parameter identification for PMSMs

Compared with observer-based estimation and compensation methods, online parameter-identification techniques provide a more intuitive approach for acquiring real-time motor parameters. By employing online identification techniques, the predictive model can be updated in real time, effectively solving the parameter-mismatch problems within the model and ensuring stable control system operation. A block diagram of the FCS-MPC online identification technique based on the PMSM is shown in Fig. 10.

Currently, the most widely applied online parameter-identification methods for PMSMs in FCS-MPC include the model reference adaptive system (MRAS), recursive least squares (RLS), extended Kalman filter (EKF), and NN algorithms.

In Ref. [84], MRAS was applied to identify inductance parameters, with fuzzy logic incorporated to optimize the traditional MRAS. This effectively reduces the current distortion and torque ripple caused by parameter mismatches, although mismatches in the resistance and flux were not addressed. In Ref. [85], the long-term RLS method was employed for online inductance identification, which, compared with conventional RLS, ensured both rapid parameter tracking and efficient noise reduction. As intelligent algorithms, such as NNs, become more prevalent, recursive NNs are also being applied to handle parameter mismatches in MPCC. In Ref. [86], a discrete-time mechanical parameter observer was introduced for the online estimation of parameter mismatches, whereas adaptive speed control was used to mitigate their impact, with the control law solved in real time using recursive NNs.

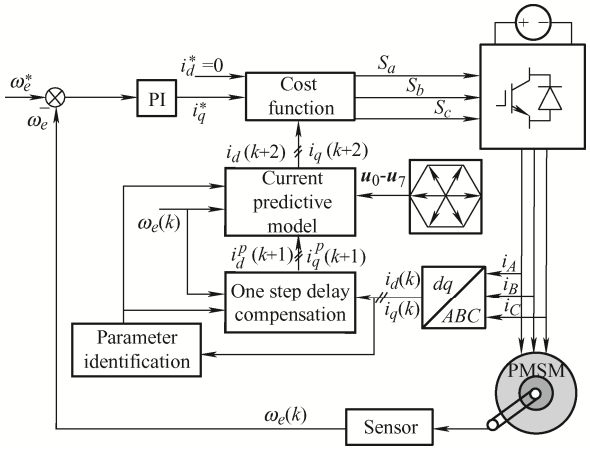


Fig. 10 Control block diagram for parameter identification based on FCS-MPCC

Although online parameter-identification techniques enable the real-time acquisition of motor parameters, the rank-deficiency problem occurs when solving for four parameters (stator resistance, *d*-*q* axis inductance, and permanent-magnet flux linkage) because the PMSM model has only two voltage balance equations. The primary approaches to address this rank deficiency are as follows: reducing the number of parameters to be identified by either reconstructing the voltage equations or assuming that certain parameters remain constant [87-89]; stepwise identification, where some parameters are identified first and then used to identify the remaining ones [90-91]; and external signal injection, in which excitation signals are injected into the *d*-axis to reformulate the mathematical model into a full-rank system [92-93].

In Ref. [87], a new current prediction model was developed along with two methods for extracting inductance information. Precise inductance data replaced the flux linkage parameter, and the study found that the impact of resistance on the prediction

accuracy was negligible, making inductance the primary factor affecting the control accuracy. This resolves the rank-deficiency problem and improves the system robustness. With respect to the stepwise identification strategy, Ref. [91] introduces a cascaded MRAS-based parameter-identification method. The first stage employs MRAS to construct a full-rank incremental equation that eliminates the influence of the flux linkage, enabling the identification of the inductance and resistance, whereas the second stage identifies the flux linkage parameters. Additionally, in Refs. [92-93], the injection of excitation signals into the *d*-axis is proposed to establish a voltage equation with $i_d \neq 0$, which, combined with the two initial voltage equations under $i_d = 0$, forms four voltage equations. Thus, a full-rank condition and the simultaneous identification of all parameters are achieved. However, injecting signals into the *d*-axis introduces additional disturbances, making this method unsuitable for control systems that require very high precision.

Tab. 7 summarizes the characteristics of the aforementioned parameter-identification methods.

Tab. 7 Summary of online parameter identification

Algorithm	Speed range	Computational burden	Benefits	Drawbacks
MRAS	Low to high	Medium	Wide speed range	Difficult to get gain values
RLS	Low to high	Medium	Easy to implement	Convergence issues
EKF	Low to high	High	Robust against system noise	High computational cost
NN	Low to high	Medium	Consider inverter non-linearity	Require extensive training time

5.3 Model optimization technique

The model optimization technique aims to minimize the number of parameters in the system model and simplify the parameter-identification process. When integrated with the two previously mentioned methods, the optimized model enhances the interference resistance of the control system. Two popular approaches in the current research are incremental model construction and the model-free predictive technique.

5.3.1 Incremental model construction method

The incremental model eliminates the flux linkage parameter by calculating the difference between two time points, leaving only the inductance and resistance parameters. This follows the same derivation as Eq. (4). From Eq. (4), the *d*-*q* axis current equation at time step *k* is derived as follows

$$\begin{cases} i_d(k) = \left(1-\dfrac{T_s R_s}{L_s}\right) i_d(k-1) + T_s \omega_e i_q(k-1) + \dfrac{T_s}{L_s} u_d(k-1) \\ i_q(k) = \left(1-\dfrac{T_s R_s}{L_s}\right) i_q(k-1) - T_s \omega_e i_d(k-1) + \dfrac{T_s}{L_s} u_q(k-1) - \dfrac{T_s \omega_e \psi_f}{L_s} \end{cases} \tag{19}$$

where $i_d(k-1)$ and $i_q(k-1)$ denote the *d*-*q* axis currents at time $k-1$, and $u_d(k-1)$ and $u_q(k-1)$ are the *d*-*q* axis voltages at time $k-1$. Subtracting Eq. (5) from Eq. (19) yields the incremental model.

$$\begin{cases} i_d(k+1) = \left(2-\dfrac{T_s R_s}{L_s}\right) i_d(k) - \left(1-\dfrac{T_s R_s}{L_s}\right) i_d(k-1) + T_s \omega_e \left[i_q(k) - i_q(k-1)\right] + \dfrac{T_s}{L_s}\left[u_d(k) - u_d(k-1)\right] \\ i_q(k+1) = \left(2-\dfrac{T_s R_s}{L_s}\right) i_q(k) - \left(1-\dfrac{T_s R_s}{L_s}\right) i_q(k-1) - T_s \omega_e \left[i_d(k) - i_d(k-1)\right] + \dfrac{T_s}{L_s}\left[u_q(k) - u_q(k-1)\right] \end{cases} \tag{20}$$

As shown in Eq. (20), constructing the incremental model removes the flux linkage parameter from the original predictive model, ensuring that flux linkage mismatches no longer adversely affect the predicted current or control system.

The incremental model is incorporated into the algorithms discussed in the previous sections to improve system robustness. In Refs. [89, 94], an incremental model was constructed to eliminate the effects of flux linkage mismatches, and an extended state observer (ESO) was used to enhance the inductance robustness by estimating and compensating for uncertainties caused by inductance mismatches. This algorithm is simple and easy to implement. In Ref. [95], the incremental model was paired with a disturbance-free deadbeat inductance observer that successfully extracted inductance information and improved the sensitivity of predictive control.

5.3.2 Model-free predictive control technique

In recent years, model-free predictive control has

gained considerable attention as an effective approach for resolving parameter-mismatch problems. This method uses the system input or output data for control, thus eliminating the influence of the system parameters. The control block diagram is shown in Fig. 11. Model-free algorithms are typically classified into two main types: the look-up table (LUT) and ultra-local model methods, both of which offer promising avenues for further research.

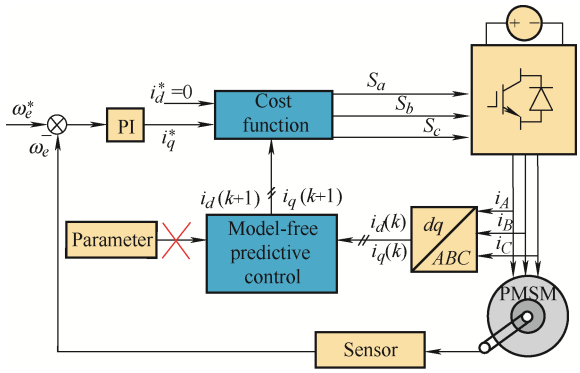


Fig. 11 Block diagram of model-free predictive control

Researchers have proposed using the current gradient produced by a voltage vector and storing it for current predictions. The PMSM current equation can be rewritten as follows

$$i_s^j(k+1)=\boldsymbol{i}_s(k)+\Delta i_s^j(k) \tag{21}$$

where $\boldsymbol{i}_s(k)=[i_d(k),\ i_q(k)]^{\mathrm{T}}$ denotes the stator d-q axis current at time step k, and $\Delta i_s^j(k)$, where $j\in\{0,1,\cdots,7\}$, represents the current gradient generated by the voltage vector in the previous cycle. Because the current gradients produced by the eight voltage vectors during each cycle can be measured, they can be stored in the LUT for use in current prediction. The current gradient updating mechanism is shown in Fig. 12 [96].

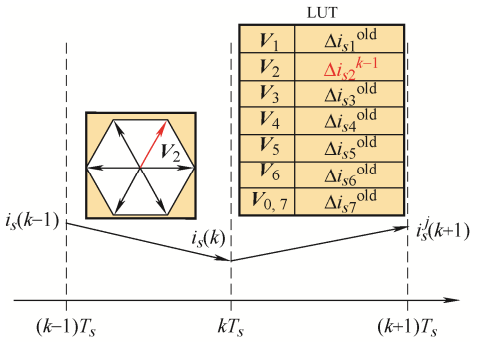


Fig. 12 Current gradient update mechanism table

As shown in Fig. 12, only one voltage vector is applied in each control cycle, and the current gradient it generates is updated in the LUT, whereas the other elements remain unchanged. If a voltage vector is not selected over several control cycles, its corresponding current gradient stagnates. Prolonged stagnation can affect the stability of the control system. To address this, Ref. [97] estimated the current gradients for all possible voltage vectors using two adjacent measured current gradients, thereby enabling the calculation of all the current gradients within a single control cycle and effectively reducing the stagnation effect. To address the stagnation problem, Ref. [98] redefined the current slope as the difference between the currents generated by two consecutively applied voltage vectors and introducing an update factor to adjust all switching states, forcing the selection of voltage vectors that have not been selected for multiple cycles.

Another approach suitable for the model-free predictive control of PMSMs is the ultra-local model method. This model is built based on system inputs and outputs, accounting for parameter uncertainties and inverter nonlinearities. With the voltage vector as the control variable and the stator current rate of change as the output variable, the first-order ultra-local model is expressed as follows [99]

$$\frac{\mathrm{d}y}{\mathrm{d}t}=F+\alpha u \tag{22}$$

where u represents the control variable, y represents the output variable, α is a scaling factor, and F comprises both the known parts of the system and the unmodeled dynamics.

Incorporating the PMSM voltage equation into the first-order ultra-local model, it can be expressed as

$$\begin{cases}\dfrac{\mathrm{d}i_d}{\mathrm{d}t}=F_d+\alpha u_d\\[2ex] \dfrac{\mathrm{d}i_q}{\mathrm{d}t}=F_q+\alpha u_q\end{cases} \tag{23}$$

where $\alpha=1/L_s$, $F_d=-R_s/L_s i_d+\omega_e i_q$, and $F_q=-R_s/L_s i_q-\omega_e i_d-1/L_s\omega_e\psi_f$.

As shown in the above equation, to eliminate the effects of parameter mismatches, the next current of the system can be accurately predicted by observing α, F_d, and F_q. Ref. [99] applied an SMO within the ultra-local model for rapid parameter observation and introduced double-vector control to enhance the current tracking performance. This approach not only

ensures system stability under parameter mismatches but also reduces current and torque ripple, offering insights for future multi-vector applications in single-cycle control. Ref. [100] developed an ultra-local model addressing parameter mismatches and inverter dead time and designed an improved ESO to estimate position disturbances, reducing the system reliance on the model and suppressing the phase current distortion caused by dead-time effects. In Ref. [101], a sliding mode controller replaced the speed-loop PI controller, and an SMO was introduced to monitor the ultra-local model. Simulation results indicated that the algorithm offers excellent robustness and strong resistance to disturbances. Ref. [102] employed model-free predictive control based on ultra-local models derived from direct speed predictive control. A control law was formulated and the inverter reference voltage was automatically generated. This approach eliminates the cascaded structure and simplifies the control system, while ensuring both stability and robustness. In Ref. [103], the ultra-local model was integrated with the RLS method to develop a model-free predictive control system for permanent-magnet motors. Additionally, the RLS method identified the inductance parameters in the proportional factor of the ultra-local model. This successful combination of the two algorithms resulted in enhanced robustness and reduced current distortion compared with conventional methods. In Ref. [104], a model-free predictive control system for a PMSM with a three-level inverter was constructed using an ultra-local model and an extended Kalman observer. The improved EKF observer ensures accurate tracking of the actual system states, enhancing the speed response of the model-free predictive control system while ensuring the precision of the current control.

Model-free predictive control for permanent-magnet motors, a promising solution for controlling system parameter mismatches, remains in the developmental phase. Challenges such as LUT update stagnation and the necessity for the real-time identification of inductance coefficients in the ultra-local model require more sophisticated theories and algorithms to achieve further improvements.

6 Future prospects and challenges

Currently, MPC demonstrates remarkable flexibility, enabling the control of different variables tailored to the specific needs of various control scenarios. This has led to its widespread use in PMSM control. However, as industrial applications continue to demand higher levels of control performance, FCS-MPC must undergo ongoing refinement. The following outlines key directions for future research.

6.1 MPC in special motors multiphase motors

With the increasing application of special-purpose motors, such as multiphase and high-speed motors, the use of FCS-MPC in special-purpose motor drive systems encounters new challenges.

In the field of multiphase motors, FCS-MPC encounters challenges, including increased computational burden [105], low-order harmonics [106-108], and regulation of the harmonic subspace.

In the field of high-speed PMSMs, control systems require fast dynamic response, high precision, and high control frequency. Consequently, higher carrier frequencies are required, leading to increased losses. In the FCS-MPC system, owing to the variable switching frequency, the average switching frequency is used in the calculations. A well-designed relationship between the switching frequency and motor speed is critical for enhancing control efficiency and reducing system losses.

Consequently, with the continuous development of microprocessors and the refinement of control algorithms, FCS-MPC systems in the field of special-purpose motors are expected to become a major research trend in the future.

6.2 Multi-step FCS-MPC strategy

Compared single-step FCS-MPC, multi-step FCS-MPC can obtain a global optimal solution. Multi-step FCS-MPC can effectively reduce the system switching frequency and improve the performance of a motor control system. However, owing to the significant increase in computational load compared with single-step prediction, it has received less attention.

To reduce the computational complexity while improving the control accuracy, many researchers have explored this topic [109-110]. This is expected to become a major research trend in multi-step FCS-MPC.

6.3 Robustness enhancement in FCS-MPC

In practical operations, PMSMs are affected by various disturbances, including external factors such as load variations, measurement errors, and unknown disturbances, as well as internal disturbances caused by parameter changes owing to current saturation, magnetic saturation, and temperature fluctuations. These disturbances degrade the performance of the control system. As discussed in Section 5, improving the accuracy of PMSM parameter identification and enhancing the effectiveness of DOs are key areas of current research and will continue to play a crucial role in the future development of FCS-MPC.

6.4 Integration with other control algorithms

As the industrial demand for control precision increases, the effectiveness of a single algorithm becomes more constrained. Combining FCS-MPC with other algorithms in PMSM control systems offers significant potential for wider applications. Depending on practical application requirements, FCS-MPC can be integrated with various control methods, such as sensorless control [110, 111-113], fuzzy control [84], sliding mode control [114-115], and fault-tolerant control [116-117], to achieve superior performance.

In summary, the analysis of future research trends in FCS-MPC for permanent-magnet motors facilitates a deeper understanding of its principles and development, offering valuable insights to further enhance the control system performance.

7 Conclusions

This paper presents a detailed analysis of the challenges in applying FCS-MPC to PMSM control systems, collates recent research findings to categorize solutions, identifies existing problems, and suggests new research directions. The major conclusions are as follows.

(1) The challenge of determining the weighting factors in the cost function. Both numerical calculation methods and weightless approaches can help mitigate the issues caused by the weighting factors.

(2) The problem of multiple vector actions in a single cycle. Analysis and derivation show that using three vectors per cycle can significantly reduce the error between the reference and output voltage vectors, although this introduces considerable computational complexity. Future studies should focus on reducing this complexity and maintaining stable switching frequencies.

(3) Improving the robustness of the control system and minimizing the impact of parameter mismatches. Simulations demonstrate the effects of parameter mismatches on *d-q* axis currents. Three approaches for addressing parameter mismatches are summarized, and the choice of method should be optimized based on specific application requirements.

The authors believe that further exploration is required to develop more effective solutions and refine the FCS-MPC algorithm.

Moreover, to further improve the FCS-MPC and expand its scope of application, this paper examines the challenges of applying FCS-MPC to multiphase permanent-magnet motors and high-speed PMSMs, laying the groundwork for future research. Integrating the FCS-MPC with other advanced control strategies, such as sensorless control, fuzzy control, and fault-tolerant control, will help meet the diverse demands of industrial applications and offer new solutions for the development of PMSM control systems.

References

[1] M Sun, Y Xu, K Han. Structure and optimization design of cup winding permanent magnet synchronous machine in flywheel energy storage system. *IEEE Transactions on Magnetics*, 2023, 59(5): 8100805.

[2] Y Yin, L Liu, S Vazquez, et al. Disturbance and uncertainty attenuation for speed regulation of PMSM servo system using adaptive optimal control strategy. *IEEE Transactions on Transportation Electrification*, 2023, 9(2): 3410-3420.

[3] M R Khowja, K Singh, A la Rocca, et al. Fault-tolerant dual channels three-phase PMSM for aerospace applications. *IEEE Access*, 2024, 12: 126845-126857.

[4] W Cui, L Ren, J Zhou, et al. A new IPMSM with hybrid rotor structure for electrical vehicle with reduced magnet

loss. *IEEE Transactions on Magnetics*, 2022, 58(2): 8700406.

[5] L Hou, J Ma, W Wang. Sliding mode predictive current control of permanent magnet synchronous motor with cascaded variable rate sliding mode speed controller. *IEEE Access*, 2022, 10: 33992-34002.

[6] G Pellegrino, A Vagati, P Guglielmi, et al. Performance comparison between surface-mounted and interior PM motor drives for electric vehicle application. *IEEE Transactions on Industrial Electronics*, 2012, 59(2): 803-811.

[7] W Xu, A K Junejo, Y Liu, et al. Improved continuous fast terminal sliding mode control with extended state observer for speed regulation of PMSM drive system. *IEEE Transactions on Vehicular Technology*, 2019, 68(11): 10465-10476.

[8] C Candelo-Zuluaga, J R Riba, A Garcia. PMSM parameter estimation for sensorless FOC based on differential power factor. *IEEE Transactions on Instrumentation and Measurement*, 2021, 70: 1504212.

[9] S G Petkar, V K Thippiripati. A novel duty-controlled DTC of a surface PMSM drive with reduced torque and flux ripples. *IEEE Transactions on Industrial Electronics*, 2023, 70(4): 3373-3383.

[10] P Cortes, M P Kazmierkowski, R M Kennel, et al. Predictive control in power electronics and drives. *IEEE Transactions on Industrial Electronics*, 2008, 55(12): 4312-4324.

[11] M M Raja, H Wang, M H Arshad, et al. Data driven model predictive control for modular multilevel converters with reduced computational complexity. *IEEE Access*, 2023, 11: 42113-42123.

[12] M Preindl, S Bolognani. Model predictive direct torque control with finite control set for PMSM drive systems, Part 1: Maximum torque per ampere operation. *IEEE Transactions on Industrial Informatics*, 2013, 9(4): 1912-1921.

[13] B Wang, J Jiao, Z Xue. Implementation of continuous control set model predictive control method for PMSM on FPGA. *IEEE Access*, 2023, 11: 12414-12425.

[14] C S Lim, S S Lee, E Levi. Continuous-control-set model predictive current control of asymmetrical six-phase drives considering system nonidealities. *IEEE Transactions on Industrial Electronics*, 2023, 70(8): 7615-7626.

[15] Z Sun, S Xu, G Ren, et al. A cascaded band based model predictive current control for PMSM drives. *IEEE Transactions on Industrial Electronics*, 2023, 70(4): 3503-3514.

[16] X Yuan, S Zhang, C Zhang, et al. Improved model predictive current control for SPMSM drives using current update mechanism. *IEEE Transactions on Industrial Electronics*, 2021, 68(3): 1938-1948.

[17] C Liu, X Xing, C Du, et al. An improved model predictive control method using optimized voltage vectors for Vienna rectifier with fixed switching frequency. *IEEE Transactions on Power Electronics*, 2023, 38(1): 358-371.

[18] A A Ahmed, B K Koh, Y I Lee. A comparison of finite control set and continuous control set model predictive control schemes for speed control of induction motors. *IEEE Transactions on Industrial Informatics*, 2018, 14(4): 1334-1346.

[19] R Fu. A simple and robust model predictive current control of PMSM using stator current predictor and target-oriented cost function. *IEEE Access*, 2022, 10: 100024-100032.

[20] J Gao, C Gong, W Li, et al. Novel compensation strategy for calculation delay of finite control set model predictive current control in PMSM. *IEEE Transactions on Industrial Electronics*, 2020, 67(7): 5816-5819.

[21] X Zhang, B Hou. Double vectors model predictive torque control without weighting factor based on voltage tracking error. *IEEE Transactions on Power Electronics*, 2018, 33(3): 2368-2380.

[22] M Gu, Y Yang, M Fan, et al. Finite control set model predictive torque control with reduced computation burden for PMSM based on discrete space vector modulation. *IEEE Transactions on Energy Conversion*, 2023, 38(1): 703-712.

[23] M Liu, K W Chan, J Hu, et al. Model predictive direct speed control with torque oscillation reduction for PMSM drives. *IEEE Transactions on Industrial Informatics*, 2019, 15(9): 4944-4956.

[24] T Wang, G Luo, Z Chen, et al. An improved robust model predictive speed control with inertia identification for PMSM drives in the electrohydrostatic actuator. *IEEE Transactions on Power Electronics*, 2023, 38(11): 13825-13841.

[25] M Norambuena, J Rodriguez, Z Zhang, et al. A very simple strategy for high-quality performance of AC machines using model predictive control. *IEEE Transactions on Power Electronics*, 2019, 34(1): 794-800.

[26] P R U Guazzelli, W C de Andrade Pereira, C M R de Oliveira, et al. Weighting factors optimization of predictive torque control of induction motor by multiobjective genetic algorithm. *IEEE Transactions on Power Electronics*, 2019, 34(7): 6628-6638.

[27] W Chen, S Zeng, G Zhang, et al. A modified double vectors model predictive torque control of permanent magnet synchronous motor. *IEEE Transactions on Power Electronics*, 2019, 34(11): 11419-11428.

[28] C Wang, J Ji, H Tang, et al. Improved model predictive current control for linear vernier permanent-magnet motor with efficient voltage vectors selection. *IEEE Transactions on Industrial Electronics*, 2023, 70(3): 2833-2842.

[29] A Brosch, S Hanke, O Wallscheid, et al. Data-driven recursive least squares estimation for model predictive current control of permanent magnet synchronous motors. *IEEE Transactions on Power Electronics*, 2021, 36(2): 2179-2190.

[30] X An, Z Liu, Q Chen, et al. Robust virtual-vector model predictive control of permanent-magnet motor considering *d-q* axis inductance parameter uncertainty. *IET Electric Power Applications*, 2024, 18(1): 76-89.

[31] J Wang, W Wang, K Atallah, et al. Demagnetization assessment for three-phase tubular brushless permanent-magnet machines. *IEEE Transactions on Magnetics*, 2008, 44(9): 2195-2203.

[32] Y Xu, N Parspour, U Vollmer. Torque ripple minimization using online estimation of the stator resistances with consideration of magnetic saturation. *IEEE Transactions on Industrial Electronics*, 2014, 61(9): 5105-5114.

[33] O Wallscheid, J Böcker. Global identification of a low-order lumped-parameter thermal network for permanent magnet synchronous motors. *IEEE Transactions on Energy Conversion*, 2016, 31(1): 354-365.

[34] R Antonello, L Ortombina, F Tinazzi, et al. Online stator resistance tracking for reluctance and interior permanent magnet synchronous motors. *IEEE Transactions on Industry Applications*, 2018, 54(4): 3405-3414.

[35] D Fernández, M Martínez, D Reigosa, et al. Impact of machine magnetization state on permanent magnet losses in permanent magnet synchronous machines. *IEEE Transactions on Industry Applications*, 2019, 55(1): 344-353.

[36] F Li, J Yue, X Ma, et al. Research on parameter identification method of permanent magnet synchronous motor. *2022 4th International Conference on Intelligent Control, Measurement and Signal Processing (ICMSP)*, Hangzhou, China. IEEE, 2022: 881-884.

[37] X Zhu, L Zhang, X Xiao, et al. Adjustable-flux permanent magnet synchronous motor sensorless drive system based on parameter-sensitive adaptive online decoupling control strategy. *IEEE Transactions on Transportation Electrification*, 2023, 9(1): 501-511.

[38] C Xie, S Zhang, X Li, et al. Parameter identification for SPMSM with deadbeat predictive current control using online PSO. *IEEE Transactions on Transportation Electrification*, 2024, 10(2): 4055-4064.

[39] Y Xu, X Ding, J Wang, et al. Three-vector-based low-complexity model predictive current control with reduced steady-state current error for permanent magnet synchronous motor. *IET Electric Power Applications*, 2020, 14(2): 305-315.

[40] Y Wang, W Liao, S Huang, et al. A robust DPCC for IPMSM based on a full parameter identification method. *IEEE Transactions on Industrial Electronics*, 2023, 70(8): 7695-7705.

[41] X Yao, S Huang, J Wang, et al. Improved ROGI-FLL-based sensorless model predictive current control with MRAS parameter identification for SPMSM drives. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 2023, 11(2): 1684-1695.

[42] Y Zhou, S Zhang, C Zhang, et al. Current prediction error based parameter identification method for SPMSM with deadbeat predictive current control. *IEEE Transactions on Energy Conversion*, 2021, 36(3): 1700-1710.

[43] C Yang, B Song, Y Xie, et al. Online parallel estimation of mechanical parameters for PMSM drives via a network of interconnected extended sliding-mode observers. *IEEE Transactions on Power Electronics*, 2021, 36(10): 11818-11834.

[44] B Wu, D Luo, M Li, et al. Parameter estimation using improved adaline neural network for sensorless control of IPMSM. *2021 IEEE 4th Student Conference on Electric Machines and Systems (SCEMS)*, Huzhou, China. IEEE, 2021: 1-6.

[45] Y Xu, X Ding, J Wang, et al. Robust three-vector-based low-complexity model predictive current control with supertwisting-algorithm-based second-order sliding-mode observer for permanent magnet synchronous motor. *IET Power Electronics*, 2019, 12(11): 2895-2903.

[46] O Babayomi, Z Zhang. Model-free predictive control of power converters with multifrequency extended state observers. *IEEE Transactions on Industrial Electronics*, 2023, 70(11): 11379-11389.

[47] M S Mousavi, S A Davari, V Nekoukar, et al. Predictive torque control of induction motor based on a robust integral sliding mode observer. *IEEE Transactions on Industrial Electronics*, 2023, 70(3): 2339-2350.

[48] M Khalilzadeh, S Vaez-Zadeh, J Rodriguez, et al. Model-free predictive control of motor drives and power converters: A review. *IEEE Access*, 2021, 9: 105733-105747.

[49] P Jiravit, S Yamamoto. A two-stage method for order selection in model-free predictive control. *IEEJ Transactions on Electrical and Electronic Engineering*, 2024, 19(6): 1035-1045.

[50] N Yang, S Zhang, X Li, et al. A new model-free deadbeat predictive current control for PMSM using parameter-free Luenberger disturbance observer. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 2023, 11(1): 407-417.

[51] X Sun, M Wu, G Lei, et al. An improved model predictive current control for PMSM drives based on current track circle. *IEEE Transactions on Industrial Electronics*, 2021, 68(5): 3782-3793.

[52] M Liu, J Hu, K W Chan, et al. Dual cost function model predictive direct speed control with duty ratio optimization for PMSM drives. *IEEE Access*, 2020, 8: 126637-126647.

[53] C Yao, Z Sun, S Xu, et al. ANN optimization of weighting factors using genetic algorithm for model predictive control of PMSM drives. *IEEE Transactions on Industry Applications*, 2022, 58(6): 7346-7362.

[54] J J Justo, F Mwasilu, E K Kim, et al. Fuzzy model predictive direct torque control of IPMSMs for electric vehicle applications. *IEEE/ASME Transactions on Mechatronics*, 2017, 22(4): 1542-1553.

[55] M Novak, H Xie, T Dragicevic, et al. Optimal cost function parameter design in predictive torque control (PTC) using artificial neural networks (ANN). *IEEE Transactions on Industrial Electronics*, 2021, 68(8): 7309-7319.

[56] F Wang, J Li, Z Li, et al. Design of model predictive control weighting factors for PMSM using Gaussian distribution-based particle swarm optimization. *IEEE Transactions on Industrial Electronics*, 2022, 69(11): 10935-10946.

[57] J Li, F Wang, D Ke, et al. Weighting factors design of model predictive control for permanent magnet synchronous machine using particle swarm optimization. *Transactions of China Electrotechnical Society*, 2021, 36(1): 50-59, 76.

[58] K Zhang, M Fan, Y Yang, et al. Tolerant sequential model predictive direct torque control of permanent magnet synchronous machine drives. *IEEE Transactions on Transportation Electrification*, 2020, 6(3): 1167-1176.

[59] X Zhang, K Yan, M Cheng. Two-stage series model predictive torque control for PMSM drives. *IEEE Transactions on Power Electronics*, 2021, 36(11): 12910-12918.

[60] X Wang, X Lin, Q Huang, et al. An improved parallel predictive torque control for permanent magnet synchronous motor. *IEEE Access*, 2023, 11: 32496-32507.

[61] Y Han, C Gong, L Yan, et al. Multiobjective finite control set model predictive control using novel delay compensation technique for PMSM. *IEEE Transactions on Power Electronics*, 2020, 35(10): 11193-11204.

[62] X Zhang, Y He. Direct voltage-selection based model predictive direct speed control for PMSM drives without weighting factor. *IEEE Transactions on Power Electronics*, 2019, 34(8): 7838-7851.

[63] A Nasr, C Gu, G Buticchi, et al. A low-complexity modulated model predictive torque and flux control strategy for PMSM drives without weighting factor. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 2023, 11(2): 1305-1316.

[64] C Ma, X Yao, H Li, et al. A novel torque boundary-based model predictive torque control for PMSM without weighting factor. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 2021, 9(4): 4395-4406.

[65] X Zhang, C Zhang, C Xu, et al. Multimode model predictive control for PMSM drive system. *IEEE Transactions on Transportation Electrification*, 2023, 9(1): 667-677.

[66] X Zhang, Z Zhao. Model predictive control for PMSM drives with variable dead-zone time. *IEEE Transactions on Power Electronics*, 2021, 36(9): 10514-10525.

[67] Y Zhang, D Xu, J Liu, et al. Performance improvement of model-predictive current control of permanent magnet synchronous motor drives. *IEEE Transactions on Industry Applications*, 2017, 53(4): 3683-3695.

[68] R Chen, K Zhai, H Shu. Predictive control of dual vector

fixed switching frequency model for permanent magnet synchronous motor. *Transactions of China Electrotechnical Society*, 2023, 38(14): 3812-3823.

[69] H Lin, S Niu, Z Xue, et al. A simplified virtual-vector-based model predictive control technique with a control factor for three-phase SPMSM drives. *IEEE Transactions on Power Electronics*, 2023, 38(6): 7546-7557.

[70] X Zhang, K Yan, W Zhang. Composite vector model predictive control with time-varying control period for PMSM drives. *IEEE Transactions on Transportation Electrification*, 2021, 7(3): 1415-1426.

[71] Z Lan, J Luo, Y Li, et al. Model prediction torque control for permanent magnet synchronous motor based on the fast selection table. *Transactions of China Electrotechnical Society*, 2023, 38(21): 5749-5757.

[72] X Wang, Z Li. Ultra-high speed motor model predictive control's double closed loop control strategy of static difference to eliminate. *Proceedings of the CSEE*, 2023, 43(22): 8897-8909.

[73] X Li, Z Xue, X Yan, et al. Voltage vector rapid screening-based three-vector model predictive torque control for permanent magnet synchronous motor. *Transactions of China Electrotechnical Society*, 2022, 37(7): 1666-1678.

[74] T Li, X Sun, G Lei, et al. Finite-control-set model predictive control of permanent magnet synchronous motor drive systems: An overview. *CAA Journal of Automatica Sinica*, 2022, 9(12): 2087-2105.

[75] H Liu, W Lin, Z Liu, et al. Model predictive current control with model-aid extended state observer compensation for PMSM drive. *IEEE Transactions on Power Electronics*, 2023, 38(3): 3152-3162.

[76] C Gong, L Ding, Y Li, et al. Flux observation transfer-based inductance identification technique for precise FCS-MPCC used in surface-mounted PMSMs. *IEEE Transactions on Power Electronics*, 2023, 38(4): 4241-4245.

[77] T Li, R Ma, W Han. Virtual-vector-based model predictive current control of five-phase PMSM with stator current and concentrated disturbance observer. *IEEE Access*, 2020, 8: 212635-212646.

[78] Y Luo, K Yang, Y Zheng. Luenberger observer-based model predictive control for six-phase PMSM motor with localization error compensation. *IEEE Transactions on Industrial Electronics*, 2023, 70(11): 10800-10810.

[79] X Guo, W Sun, D Jiang, et al. Application of the Luenberger observer for rotor resistance estimation in induction motor drives. *2023 IEEE 6th International Electrical and Energy Conference (CIEEC)*, Hefei, China. IEEE, 2023: 3254-3259.

[80] X Liu, W Li, Z Chen, et al. Improved model predictive current control of PMSM based on luenberger observer. *2021 China Automation Congress (CAC)*, Beijing, China. IEEE, 2021: 3084-3089.

[81] F Yang, X Zhao, H Jin, et al. Improved deadbeat predictive current control with embedded resonant polynomial and disturbance observer for PMSM current distortion rejection. *IEEE Journal of Emerging and Selected Topics in Power Electronics*, 2024, 12(2): 1934-1945.

[82] M Wei, C Zhang, X Qiao. An enhanced deadbeat model predictive flux control for PMSMs with a reduced order disturbance observer. *2023 IEEE International Conference on Predictive Control of Electrical Drives and Power Electronics (PRECEDE)*, Wuhan, China. IEEE, 2023: 1-6.

[83] F Yang, K Yang, Y Zhang, et al. Model predictive direct speed control of permanent magnet synchronous motor based on full parameters disturbance observer. *2020 23rd International Conference on Electrical Machines and Systems (ICEMS)*, Hamamatsu, Japan. IEEE, 2020: 1391-1396.

[84] X An, G Liu, Q Chen, et al. Adjustable model predictive control for IPMSM drives based on online stator inductance identification. *IEEE Transactions on Industrial Electronics*, 2022, 69(4): 3368-3381.

[85] A Brosch, O Wallscheid, J Bocker. Long-term memory recursive least squares online identification of highly utilized permanent magnet synchronous motors for finite-control-set model predictive control. *IEEE Transactions on Power Electronics*, 2023, 38(2): 1451-1467.

[86] T T Nguyen, H N Tran, T H Nguyen, et al. Recurrent neural network-based robust adaptive model predictive speed control for PMSM with parameter mismatch. *IEEE Transactions on Industrial Electronics*, 2023, 70(6): 6219-6228.

[87] X Zhang, Z Wang. Simple robust model predictive current control for PMSM drives without flux-linkage parameter. *IEEE Transactions on Industrial Electronics*, 2023, 70(4): 3515-3524.

[88] X Zhang, Z Wang, Z Zhao, et al. Model predictive voltage control for SPMSM drives with parameter robustness optimization. *IEEE Transactions on Transportation Electrification*, 2022, 8(3): 3151-3163.

[89] X Zhang, L Zhang, Y Zhang. Model predictive current control for PMSM drives with parameter robustness improvement. *IEEE Transactions on Power Electronics*, 2019, 34(2): 1645-1657.

[90] M S Rafaq, F Mwasilu, J Kim, et al. Online parameter identification for model-based sensorless control of interior permanent magnet synchronous machine. *IEEE Transactions on Power Electronics*, 2017, 32(6): 4631-4643.

[91] X Yao, S Huang, J Wang, et al. A two-vector-based model predictive current control with online parameter identification for PMSM drives. *Proceedings of the CSEE*, 2023, 43(23): 9319-9329.

[92] Q Wang, G Wang, N Zhao, et al. An impedance model-based multiparameter identification method of PMSM for both offline and online conditions. *IEEE Transactions on Power Electronics*, 2021, 36(1): 727-738.

[93] Y Yao, Y Huang, F Peng, et al. An improved deadbeat predictive current control with online parameter identification for surface-mounted PMSMs. *IEEE Transactions on Industrial Electronics*, 2020, 67(12): 10145-10155.

[94] M Yang, X Lang, J Long, et al. Flux immunity robust predictive current control with incremental model and extended state observer for PMSM drive. *IEEE Transactions on Power Electronics*, 2017, 32(12): 9267-9279.

[95] X Zhang, Y Cheng, L Zhang. Disturbance-deadbeat inductance observer-based current predictive control for surface-mounted permanent magnet synchronous motors drives. *IET Power Electronics*, 2020, 13(6): 1172-1180.

[96] C K Lin, T H Liu, J T Yu, et al. Model-free predictive current control for interior permanent-magnet synchronous motor drives based on current difference detection technique. *IEEE Transactions on Industrial Electronics*, 2014, 61(2): 667-681.

[97] C Ma, H Li, X Yao, et al. An improved model-free predictive current control with advanced current gradient updating mechanism. *IEEE Transactions on Industrial Electronics*, 2021, 68(12): 11968-11979.

[98] C A Agustin, J T Yu, Y S Cheng, et al. A synchronized current difference updating technique for model-free predictive current control of PMSM drives. *IEEE Access*, 2021, 9: 63306-63318.

[99] Z Sun, Y Deng, J Wang, et al. Finite control set model-free predictive current control of PMSM with two voltage vectors based on ultralocal model. *IEEE Transactions on Power Electronics*, 2023, 38(1): 776-788.

[100] D Wen, Y Zhang, Y Zhang. Three-vector model-free predictive control for permanent magnet synchronous motor. *IET Power Electronics*, 2023, 16(16): 2754-2768.

[101] Y Zhang, L Shao, Z Yin, et al. Finite control set model free predictive current control of permanent magnet synchronous motors. *2023 IEEE International Conference on Predictive Control of Electrical Drives and Power Electronics (PRECEDE)*, Wuhan, China. IEEE, 2023: 1-6.

[102] J Mao, H Li, L Yang, et al. Non-cascaded model-free predictive speed control of SMPMSM drive system. *IEEE Transactions on Energy Conversion*, 2022, 37(1): 153-162.

[103] Y Wei, D Ke, D Huang, et al. Model-free predictive sliding mode control using ultra-localized time-series for permanent magnet synchronous motor drives. *Transactions of China Electrotechnical Society*, 2024, 39(4): 1022-1032.

[104] F Wang, Z Ke, D Ke, et al. Model-free predictive current control of three-level inverter-fed PMSM based on strong tracking extended Kalman observer. *Proceedings of the CSEE*, 2023, 43(22): 8910-8921.

[105] X Song, H Wang, X Ma, et al. Robust model predictive current control for a nine-phase open-end winding PMSM with high computational efficiency. *IEEE Transactions on Power Electronics*, 2023, 38(11): 13933-13943.

[106] C Xue, W Song, X Wu, et al. Finite-control-set model predictive torque control algorithm with deadbeat optimization for five-phase permanent-magnet synchronous machines drives. *Proceedings of the CSEE*, 2017, 37(23): 7014-7023.

[107] W Song, Y Guo, B Yu, et al. A model predictive current control of five-phase PMSM with duty ratio optimization in whole speed range. *Proceedings of the CSEE*, 2022, 42(12): 4561-4570.

[108] W Tao, H Zhou, C Chen, et al. Novel virtual-finite-control-set-based model predictive torque control of five-phase PMSM with enhanced third harmonic current suppression. *IEEE Journal of*

Emerging and Selected Topics in Power Electronics, 2024, 12(3): 2820-2830.

[109] X Li, W Tian, Q Yang, et al. Augmented multistep finite-control-set model predictive control for induction motor-drive system. *IEEE Transactions on Power Electronics*, 2023, 38(11): 13842-13854.

[110] W Chen, X Zhang, X Gu, et al. Band-based multi-step predictive torque control strategy for PMSM drives. *IEEE Access*, 2019, 7: 171411-171422.

[111] R Sreejith, B Singh. Sensorless predictive control of SPMSM-driven light EV drive using modified speed adaptive super twisting sliding mode observer with MAF-PLL. *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, 2021, 2(1): 42-52.

[112] P Madevi, P D Kumar, T Ramesh. Speed sensorless control of model predictive current control based two-level inverter fed five phase permanent magnet synchronous motor drive. *2023 IEEE IAS Global Conference on Emerging Technologies (GlobConET)*, London, UK. IEEE, 2023: 1-6.

[113] X Sun, T Li, Z Zhu, et al. Speed sensorless model predictive current control based on finite position set for PMSHM drives. *IEEE Transactions on Transportation Electrification*, 2021, 7(4): 2743-2752.

[114] C Liu, H Liu, L Han, et al. Multi-level coordinated yaw stability control based on sliding mode predictive control for distributed drive electric vehicles under extreme conditions. *IEEE Transactions on Vehicular Technology*, 2023, 72(1): 280-296.

[115] X Liu, H Yang, H Lin, et al. A novel finite-set sliding-mode model-free predictive current control for PMSM drives without DC-link voltage sensor. *IEEE Transactions on Power Electronics*, 2024, 39(1): 320-331.

[116] L Feng, Z Wang, J Feng, et al. Cascaded model predictive control of six-phase permanent magnet synchronous motor with fault tolerant ability. *CES Transactions on Electrical Machines and Systems*, 2023, 7(3): 311-319.

[117] S Cheng, R Zhou, Z Li, et al. Robust model-free fault-tolerant predictive control for PMSM drive system. *IEEE Access*, 2024, 12: 8502-8512.

Bo Wang received B.S. and M.S. degrees in Electrical Engineering from Shenyang University of Technology, Shenyang, China, in 2017 and 2021, respectively. He is currently working toward a Ph.D. degree in Electrical Engineering at Shenyang University of Technology, Shenyang, China.

His current research interests include model predictive control and sensorless control.

Shi Jin received B.S., M.S., and Ph.D. degrees in Electrical Engineering from Shenyang University of Technology, Shenyang, China, in 2004, 2007, and 2011, respectively. She is currently a Professor with the School of Electrical Engineering, Shenyang University of Technology, Shenyang, China.

Her major research interests and activities primarily include power electronics technology, electrical machines and their control systems, and wind power generation.

Peng Sun received B.S. and M.S. degrees in Electrical Engineering from Shenyang University of Technology, Shenyang, China, in 2015 and 2018, respectively. He is currently working toward a Ph.D. degree in Electrical Engineering at Shenyang University of Technology.

His current research interests lie in high-speed PMSM control.

Guangwei Liu received the B.E., M.S., and Ph.D. degrees in Electrical Engineering from the Shenyang University of Technology, Shenyang, China, in 2005, 2008, and 2015, respectively. He is currently a Professor with the School of Electrical Engineering, Shenyang University of Technology, Shenyang, China.

His major research interests and activities are in the area of permanent magnet electrical machines and their drive systems.